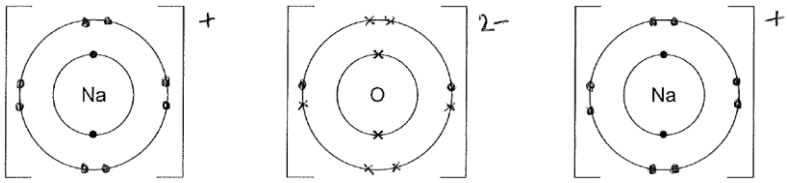


HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		 award (1) for 8 dots in outer shell of both sodium ions award (1) for 6 crosses and 2 dots in outer shell of oxide ion do not accept if extra electrons drawn in inner shell award (1) for + charge on sodium ions and 2– charge on oxide ion accept if + charge shown on only one sodium ion		3		3		
		(ii)		$4\text{Na} + \text{O}_2 \longrightarrow \boxed{2} \text{Na}_2\text{O}$ award (1) for product award (1) for balancing only if product correct		2		2	1	